UNIT 3

Theory of Elasticity, Westergaard Solution (Mode I) & the K–G Relation

stress & strain review | Airy function | Westergaard | stresses & displacements | $G = K^2/E'$

1. Review of Theory of Elasticity

LEFM is built on the **linear theory of elasticity**; the crack-tip stresses are obtained by solving the elasticity equations with the crack-face boundary conditions.

1.1 Stress and equilibrium

- Stress tensor σ_{ij} $(i, j = x, y, z)$: 3 normal $(\sigma_x, \sigma_y, \sigma_z)$ + 3 shear $(\tau_{xy} = \tau_{yx}, \tau_{yz}, \tau_{zx})$ components.
- **Equilibrium** (no body force), 2-D:

$$\partial\sigma_x/\partial x + \partial\tau_{xy}/\partial y = 0 \; ; \; \partial\tau_{xy}/\partial x + \partial\sigma_y/\partial y = 0$$

- Principal stresses (2-D): $\sigma_{1,2} = (\sigma_x + \sigma_y)/2 \pm \sqrt{[\,((\sigma_x - \sigma_y)/2)^2 + \tau_{xy}^2\,]}$; max shear $\tau_{max} = (\sigma_1 - \sigma_2)/2$.

1.2 Strain & compatibility

$$\varepsilon_x = \partial u/\partial x \quad \varepsilon_y = \partial v/\partial y \quad \gamma_{xy} = \partial u/\partial y + \partial v/\partial x$$

- u, v = displacements in x, y. For a single-valued continuous displacement field the strains must satisfy **compatibility**:

$$\partial^2\varepsilon_x/\partial y^2 + \partial^2\varepsilon_y/\partial x^2 = \partial^2\gamma_{xy}/\partial x\partial y$$

1.3 Hooke's law, plane stress & plane strain

- Isotropic: $\varepsilon_x = [\sigma_x - \nu(\sigma_y + \sigma_z)]/E$; $\varepsilon_y = [\sigma_y - \nu(\sigma_x + \sigma_z)]/E$; $\gamma_{xy} = \tau_{xy}/\mu$; shear modulus $\mu = E / 2(1 + \nu)$.

	Plane stress (thin plate)	Plane strain (thick plate)
Assumption	$\sigma_z = \tau_{xz} = \tau_{yz} = 0$	$\varepsilon_z = \gamma_{xz} = \gamma_{yz} = 0$
Out-of-plane	$\varepsilon_z = -\nu(\sigma_x + \sigma_y)/E$	$\sigma_z = \nu(\sigma_x + \sigma_y)$
Kolosov constant κ	$(3 - \nu)/(1 + \nu)$	$3 - 4\nu$
Modulus E'	E	$E/(1 - \nu^2)$

1.4 Airy stress function φ(x, y)

Choose a scalar φ that **automatically satisfies equilibrium**:

$$\sigma_x = \partial^2\varphi/\partial y^2 \quad \sigma_y = \partial^2\varphi/\partial x^2 \quad \tau_{xy} = -\partial^2\varphi/\partial x\partial y$$

Substituting in compatibility (with Hooke's law) gives the **bi-harmonic equation**:

$$\nabla^4\varphi = \partial^4\varphi/\partial x^4 + 2\,\partial^4\varphi/\partial x^2\partial y^2 + \partial^4\varphi/\partial y^4 = 0$$

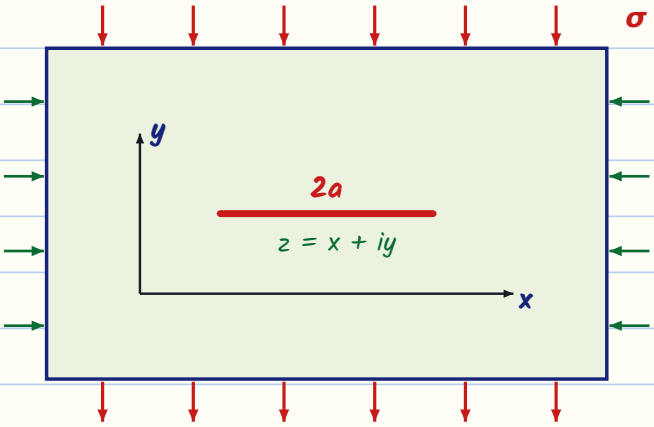
same for plane stress & strain

- Solve $\nabla^4\varphi = 0$ with the boundary conditions → stresses; strains from Hooke; displacements by integration.
- In polar form: $\sigma_r = (1/r)\,\partial\varphi/\partial r + (1/r^2)\,\partial^2\varphi/\partial\theta^2$, $\sigma_\theta = \partial^2\varphi/\partial r^2$, $\tau_r\theta = -\partial/\partial r[(1/r)\,\partial\varphi/\partial\theta]$.

1.5 Complex-variable method

- Let $z = x + iy = r\,e^{i\theta}$. A function $f(z) = \alpha + i\beta$ is **analytic** if Cauchy–Riemann conditions $\partial\alpha/\partial x = \partial\beta/\partial y$, $\partial\alpha/\partial y = -\partial\beta/\partial x$ hold. Then α and β are harmonic: $\nabla^2\alpha = \nabla^2\beta = 0$.
- Hence $\varphi = \mathrm{Re}[\,\mathrm{conj}(z)\,\psi(z) + \chi(z)\,]$ (Goursat) satisfies $\nabla^4\varphi = 0$ — any two analytic functions give a valid stress field (Kolosov–Muskhelishvili). Westergaard (1939) simplified this to **one** function Z(z) for symmetric (Mode I) problems.

2. Westergaard Solution – Mode I



Centre crack, infinite plate, biaxial remote stress σ

$$Z_I(z) = \sigma\, z \,/\, \sqrt{(z^2 - a^2)}$$

B.C.: $\sigma_y = 0$ on $|x| < a$

$\sigma_x = \sigma_y = \sigma$ as $|z| \to \infty$

Fig 3.1 Westergaard problem: crack 2a in an infinite plate with biaxial σ

Westergaard stress function (Mode I)

> Airy function: $\varphi = \mathrm{Re}\, Z_2 + y\, \mathrm{Im}\, Z_1$ where $Z_1 = \int Z\, dz$, $Z_2 = \int Z_1\, dz$ and $Z(z)$ is analytic.

> Stresses: $\sigma_x = \mathrm{Re}\, Z - y\, \mathrm{Im}\, Z'$ $\sigma_y = \mathrm{Re}\, Z + y\, \mathrm{Im}\, Z'$ $\tau_{xy} = -y\, \mathrm{Re}\, Z'$ ($Z' = dZ/dz$)

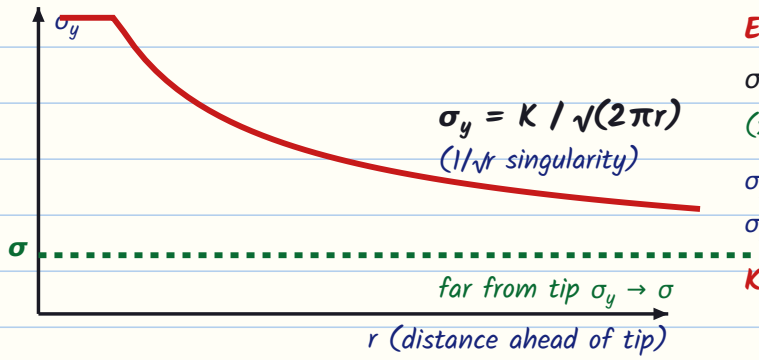
> Displacements (2μ factor): $2\mu u = ((\kappa-1)/2)\, \mathrm{Re}\, Z_1 - y\, \mathrm{Im}\, Z$; $2\mu v = ((\kappa+1)/2)\, \mathrm{Im}\, Z_1 - y\, \mathrm{Re}\, Z$

2.1 Check of boundary conditions for a crack

For a crack on $-a \le x \le a$ ($y = 0$) with remote biaxial tension σ, take

$$Z_1(z) = \sigma z / \sqrt{(z^2 - a^2)}$$

1. Far field ($|z| \to \infty$): $Z \to \sigma$, $Z' \to 0 \Rightarrow \sigma_x = \sigma_y = \sigma$, $\tau_{xy} = 0$ ✔.
2. On crack faces $y = 0$, $|x| < a$: $\sqrt{(x^2 - a^2)}$ is imaginary $\Rightarrow Z$ is purely imaginary $\Rightarrow \sigma_y = \mathrm{Re}\, Z = 0$ and $\tau_{xy} = 0$ ✔ (traction-free).
3. Ahead of crack $y = 0$, $x > a$: Z real $\Rightarrow$ **$\sigma_y = \sigma x / \sqrt{(x^2 - a^2)}$** $\to$ singular at $x = a$.



Exact (Westergaard):

$\sigma_y = \sigma x / \sqrt{(x^2 - a^2)}$

(x = distance from crack centre)

$\sigma_y \to \infty$ as $x \to a$: singular

$\sigma_y \to \sigma$ as $x \to \infty$

K-field valid for $r \ll a$

Fig 3.2 Stress ahead of the crack tip ($\theta = 0$)

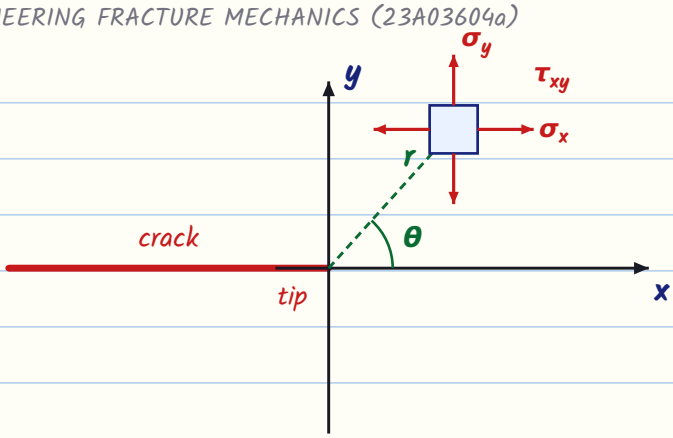
Remark (uniaxial loading): if $\sigma_x = 0$ at infinity the above solution must be corrected by a uniform $\sigma_x = -\sigma$ (**T-stress**). It does not change σ_y, τ_{xy} or K, only σ_x (matters in Unit IV).

2.2 Stress field near the tip – K defined

Shift the origin to the right tip: $z = a + \zeta$, $\zeta = r e^{i\theta}$, $r \ll a$. Then $z^2 - a^2 = \zeta(2a + \zeta) \approx 2a\zeta$ and

$$Z_1 \approx \sigma a / \sqrt{(2a\zeta)} = \sigma\sqrt{(\pi a)} / \sqrt{(2\pi\zeta)} = K_I / \sqrt{(2\pi\zeta)}$$

$K_I = \sigma\sqrt{(\pi a)}$



Polar coordinates at the tip:

$x = r\cos\theta$, $y = r\sin\theta$

$\theta = 0$: line ahead of the crack

$\theta = \pm\pi$: crack faces (free)

Origin shifted from crack centre to the crack tip: $z = a + \zeta$

$\zeta = r\,e^{i\theta}$

Fig 3.3 Crack-tip coordinate system and stress components

Differentiate & separate real and imaginary parts: $Z = K/\sqrt{(2\pi r)} \cdot e^{-i\theta/2}$, $Z' = -K/(2\sqrt{(2\pi)}\, r^{3/2}) \cdot e^{-3i\theta/2}$. Using $\sigma_y = \mathrm{Re}\, Z + y\, \mathrm{Im}\, Z'$ etc. ($y = r\sin\theta$):

$$\sigma_x = K_I/\sqrt{(2\pi r)} \cdot \cos(\theta/2)\,[\,1 - \sin(\theta/2)\sin(3\theta/2)\,]$$

$$\sigma_y = K_I/\sqrt{(2\pi r)} \cdot \cos(\theta/2)\,[\,1 + \sin(\theta/2)\sin(3\theta/2)\,]$$

$$\tau_{xy} = K_I/\sqrt{(2\pi r)} \cdot \cos(\theta/2)\sin(\theta/2)\cos(3\theta/2)$$

Plane strain out-of-plane stress

> $\sigma_z = \nu\,(\sigma_x + \sigma_y) = 2\,\nu\,K_I\cos(\theta/2) / \sqrt{(2\pi r)}$ (plane strain); $\sigma_z = 0$ (plane stress).

> Principal stresses: $\sigma_{1,2} = K/\sqrt{(2\pi r)}\cos(\theta/2)\,[1 \pm \sin(\theta/2)]$; $\sigma_3 = \nu(\sigma_1+\sigma_2)$ (plane strain).

2.3 Properties of the K-field

- **Singularity** $\propto r^{-1/2}$: stresses $\to \infty$ as $r \to 0$; only the *intensity* K is finite. (Real material yields at the tip – plastic zone.)
- On the crack line $\theta = 0$: $\sigma_x = \sigma_y = K/\sqrt{(2\pi r)}$, $\tau_{xy} = 0$ (pure hydrostatic-like tension, max triaxiality in plane strain).
- **Definition:** $K_I = \lim_{r\to 0} \sigma_y(r, \theta = 0)\ \sqrt{(2\pi r)}$. Unit: MPa$\sqrt{m}$ (or N/m$^{3/2}$).
- The shape of the field ($f_{ij}(\theta)$) is universal; **K scales its magnitude** – same K ⇒ same crack-tip state for any geometry (basis of similitude, K_{Ic} transferable from specimen to structure).
- K depends linearly on load, and on crack size & geometry: **$K = \sigma\, Y\, \sqrt{(\pi a)}$**. Superposition of K's valid for the **same mode**.
- K-dominance zone: valid for r small compared with a, ligament and thickness ($r <$ ~ a/10 to a/50); beyond that, higher-order terms/T-stress matter.

Worked Example 1 (stresses near the tip)

K_I = 40 MPa√m, plane strain, ν = 0.3. At r = 2 mm find σ_x, σ_y, τ_{xy}, σ_z at θ = 0 and θ = 90°.

$K/\sqrt{(2\pi r)} = 40/\sqrt{(2\pi\times0.002)} = 40/0.1121 =$ **356.8 MPa**.

θ = 0 : $\sigma_x = \sigma_y =$ **356.8 MPa**, $\tau_{xy} = 0$, $\sigma_z = \nu(\sigma_x+\sigma_y) = 0.3\times713.6 =$ **214 MPa**.

θ = 90° : cos45° = sin45° = 0.7071, sin135° = 0.7071, cos135° = −0.7071 → σ_x = 356.8×0.7071×(1−0.5) = **126 MPa** ; σ_y = 356.8×0.7071×1.5 = **378 MPa** ; τ_{xy} = 356.8×0.5×(−0.7071) = **−126 MPa** ; σ_z = 0.3×(126+378) = **151 MPa**.

3. Displacements and Crack Opening

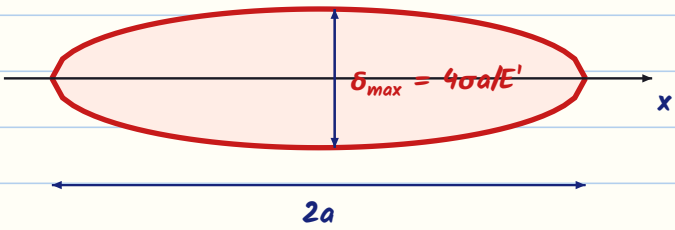
Integrating Z: $Z_I = \int Z\, d\zeta = K\sqrt{(2\zeta/\pi)}$. Then near the tip (κ = Kolosov constant):

$$u = (K_I/2\mu)\sqrt{(r/2\pi)}\cos(\theta/2)\,[\,\kappa - 1 + 2\sin^2(\theta/2)\,]$$

$$v = (K_I/2\mu)\sqrt{(r/2\pi)}\sin(\theta/2)\,[\,\kappa + 1 - 2\cos^2(\theta/2)\,]$$

- Displacements ∝ √r (finite at tip, strain energy finite). On the crack faces θ = ±π: **$v = \pm(\kappa+1)K_I/(2\mu)\cdot\sqrt{(r/2\pi)}$** , $u = \pm(\kappa-1)K_I/(2\mu)\cdot\sqrt{(r/2\pi)}$.
- Full-field: using $Z_I = \sigma\sqrt{(z^2-a^2)}$ the face opening of the Griffith crack is an **ellipse**:

$$v(x) = (2\sigma/E')\sqrt{(a^2-x^2)} \Rightarrow \delta_{max}\ (\text{centre, total}) = 4\sigma a/E'$$



Crack-face displacement:

$v(x) = (2\sigma/E')\sqrt{(a^2-x^2)}$

(each face, using $\int Z\,dz = \sigma\sqrt{(z^2-a^2)}$)

Shape = ELLIPSE (not a wedge)

$E' = E$ (plane stress)

$E' = E/(1-\nu^2)$ (plane strain)

Fig 3.4 Crack opening displacement profile (elliptical)

- CTOD concept: opening at distance r behind the tip $\delta = 2v = (\kappa+1)K/\mu\cdot\sqrt{(r/2\pi)}$ → used by Irwin to define δ_t (Unit IV).
- Strain energy density near tip ∝ 1/r (integrable singularity) – the basis of Sih's S-criterion.

Worked Example 2 (crack face opening)

K_I = 30 MPa√m, E = 70 GPa, ν = 0.33 (plane strain). Find v at r = 1 mm behind tip on the upper face.

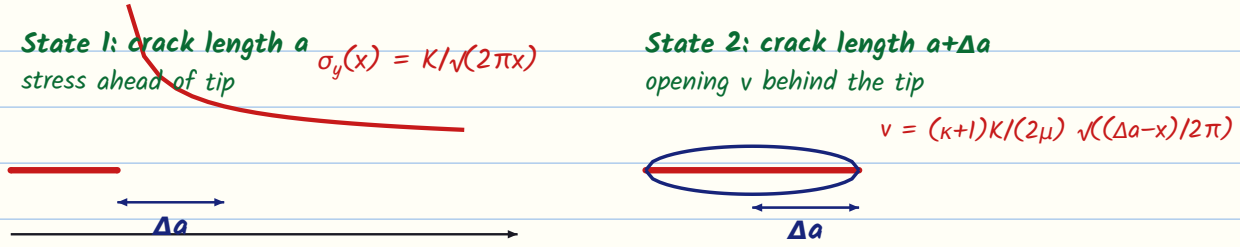
$\mu = E/2(1+\nu)$ = 26.3 GPa ; $\kappa = 3 - 4\nu$ = 1.68 ; $v = (K/2\mu)\sqrt{(r/2\pi)}(\kappa+1)$ = $(30\times10^6/(2\times26.3\times10^9))\times\sqrt{(10^{-3}/2\pi)}\times2.68$ = 5.70×10^{-4} × 0.01262 × 2.68 = **1.93×10^{-5} m ≈ 19.3 µm**.

Worked Example 3 (central crack)

Steel plate (E = 210 GPa, plane stress), $2a$ = 40 mm, σ = 150 MPa. $K = \sigma\sqrt{(\pi a)}$ = $150\sqrt{(\pi\times0.02)}$ = **37.6 MPa√m**; centre opening $\delta_{max} = 4\sigma a/E$ = $4\times150\times10^6\times0.02/(210\times10^9)$ = **57 µm**.

4. Relationship Between K and G

Irwin's crack closure integral: G equals the work needed to close the crack by a small length Δa (= work released when it extends by Δa). Consider two states: (1) crack a with stress $\sigma_y(x)$ ahead of tip; (2) crack $a + \Delta a$ with opening v behind the new tip.



$$G = \lim (1/\Delta a) \int_0^{\Delta a} \sigma_y(x) \cdot v(\Delta a - x)\, dx = \text{work to close the crack by } \Delta a$$

Fig 3.5 Irwin's crack-closure integral

Work to close = ½ ∫ $\sigma_y \cdot (2v)\, dx$ = ∫ $\sigma_y\, v\, dx$ (upper + lower face, linear spring, factor ½):

$$G = \lim_{\Delta a \to 0} (1/\Delta a) \int_0^{\Delta a} \sigma_y(x)\, v(\Delta a - x)\, dx$$

Substitute $\sigma_y = K/\sqrt{(2\pi x)}$ and $v(\Delta a - x) = (\kappa+1)\, K/(2\mu) \cdot \sqrt{((\Delta a - x)/2\pi)}$:

$$G = (\kappa+1)\, K^2 / (4\pi\, \mu\, \Delta a) \int_0^{\Delta a} \sqrt{((\Delta a - x)/x)}\, dx = (\kappa+1)\, K^2 / (4\pi\, \mu\, \Delta a) \cdot (\pi\, \Delta a / 2)$$

$$G = (\kappa + 1)\, K^2 / (8\, \mu)$$

- Plane stress: $\kappa + 1 = 4/(1+\nu)$, $\mu = E/2(1+\nu) \Rightarrow G = K^2/E$.
- Plane strain: $\kappa + 1 = 4(1-\nu)$, $\Rightarrow G = (1-\nu^2)\, K^2/E$.

K – G relation (very important)

> $G = K_I^2 / E'$ with $E' = E$ (plane stress) ; $E' = E/(1-\nu^2)$ (plane strain).

> Mixed mode (I, II, III): $G = (K_I^2 + K_{II}^2)/E' + K_{III}^2/(2\mu)$.

> Fracture toughness link: $K_{Ic} = \sqrt{(E' G_{Ic})}$; $G_c = K_c^2/E'$.

> Check with Griffith: $K = \sigma\sqrt{(\pi a)} \Rightarrow G = \pi \sigma^2 a/E$ ✓ (same as the energy approach).

> ∴ Energy criterion ($G = G_c$) and stress-field criterion ($K = K_c$) are **equivalent** in LEFM.

Closure integral also shows G is **always positive** and additive only between modes: $G_{total} = G_I + G_{II} + G_{III}$ (no cross terms because work done by σ_y on v_{II} etc. is zero).

Worked Example 4

Crack $2a = 50$ mm in a wide plate under $\sigma = 100$ MPa ($E = 70$ GPa, plane stress). $K = 100\sqrt{(\pi \times 0.025)} = 28.0$ **MPa√m** ; $G = K^2/E = (28.0\times10^6)^2/(70\times10^9) = 1.12\times10^4$ **J/m² = 11.2 kJ/m²**.

Worked Example 5

Steel $K_{Ic} = 50$ MPa√m, $E = 210$ GPa, $\nu = 0.3$. Plane-strain $G_{Ic} = (1-\nu^2)K_{Ic}^2/E = 0.91\times(50\times10^6)^2/(210\times10^9) =$ **10.8 kJ/m²**. For $\sigma = 250$ MPa critical half-crack: $a_c = (1/\pi)(K_{Ic}/\sigma)^2 = (1/\pi)(0.2)^2 =$ **12.7 mm**.

Worked Example 6 (ratio of toughness)

Plane-stress $K_c = 100$ MPa√m in thin sheet vs plane-strain $K_{Ic} = 40$ MPa√m of the same alloy ⇒ G_c ratio ≈ $(100/40)^2/(1-\nu^2)$ ≈ 6.9 (E same): thin sheet absorbs ≈ 7 × more energy per unit area.

4A. Applications of the Westergaard Method

Different analytic $Z_I(z)$ give K for different **Mode I** loadings of a crack in an infinite plate:

Loading on crack	Westergaard function / result
Remote biaxial σ	$Z = \sigma z/\sqrt{(z^2-a^2)} \Rightarrow K = \sigma\sqrt{(\pi a)}$
Uniform pressure p on crack faces	same as remote $\sigma = p \Rightarrow$ **$K = p\sqrt{(\pi a)}$**
Pair of splitting forces P (per unit thickness) at x = b	$K_{-\pm} = (P/\sqrt{(\pi a)})\ \sqrt{[(a \pm b)/(a \mp b)]}$; at centre (b = 0): **$K = P/\sqrt{(\pi a)}$**
Periodic array of collinear cracks (2a, pitch 2W)	$K = \sigma\sqrt{(\pi a)} \cdot \sqrt{[(2W/\pi a)\tan(\pi a/2W)]}$
Finite width strip (Feddersen / secant)	$K = \sigma\sqrt{(\pi a)}\ \sqrt{(\sec(\pi a/W))}$ (W = width, 2a crack)

- For P at the **centre** the point load is carried to the tips; K falls as the crack grows (the

wedge-force effect), so growth is stable in wedge-loaded specimens – used in DCB / wedge-opening-load (WOL) tests.

- Collinear array: interaction between neighbouring tips raises K above $\sigma\sqrt{(\pi a)}$ (factor → 1 as a/W → 0 and → ∞ as 2a → pitch).
- **Limitations of the original Westergaard method:** (i) only symmetric (Mode I) problems with $\sigma_x = \sigma_y$ at infinity; for $\sigma_x \neq \sigma_y$ one adds a constant (Eftis–Liebowitz): $\sigma_x = \text{Re } Z - y \text{ Im } Z' - A$, where A = difference of remote stresses — the **T-stress** term; (ii) boundary conditions satisfied only at the crack faces and infinity – finite geometry corrections need other methods.

Mode II and III (preview of Unit IV)

- Mode II: analogous complex function Z_{II} gives $K_{II} = \tau\sqrt{(\pi a)}$ and $\sigma_{ij} = K_{II}/\sqrt{(2\pi r)}\ f_{ij}^{II}(\theta)$.
- Mode III: displacement w(x, y) is harmonic ⇒ τ_{xz}, τ_{yz} from an analytic function; $K_{III} = \tau\sqrt{(\pi a)}$.

4B. How is K Determined?

Method	Principle
Analytical (Westergaard, Muskhelishvili)	Closed form for simple geometries (infinite plate); handbooks (Tada, Rooke, Murakami)
Compliance method	Measure C(a); $G = (P^2/2B)\ dC/da$; $K = \sqrt{(E'G)}$
Displacement extrapolation (FEM / DIC)	$K = [2\mu/(\kappa+1)]\ \sqrt{(2\pi/r)} \cdot v(r, \theta=\pi)$, $r \to 0$
Stress extrapolation	$K = \lim \sigma_y \sqrt{(2\pi r)}$ as $r \to 0$, $\theta = 0$
Virtual crack closure (VCCT)	$G_I \approx F_y \Delta v / (2B\Delta a)$ from nodal force & opening
J-integral (domain integral)	$J = G \Rightarrow K$; path independent – most accurate in FEM
Photoelasticity / caustics / strain gauges	Isochromatic fringes, shadow-spot (caustic) diameter, gauges at known (r, θ) near tip

- FEM uses **quarter-point (Barsoum / Henshell–Shaw) singular elements**: mid-side nodes shifted to ¼ of the side from the tip give a $1/\sqrt{r}$ strain singularity.
- Check with Example 2: $K = 2\mu\ v/(\kappa+1) \cdot \sqrt{(2\pi/r)} = 2\times26.3\times10^{9}\times1.93\times10^{-5}/2.68 \times \sqrt{(2\pi/10^{-3})} \approx 30$ MPa√m ✔.

4C. More on Elasticity Basics

A. Stress transformation & Mohr's circle

$$\sigma_x' = (\sigma_x+\sigma_y)/2 + (\sigma_x-\sigma_y)/2 \cdot \cos 2\alpha + \tau_{xy} \sin 2\alpha$$

$$\tau_{x'y'} = -(\sigma_x-\sigma_y)/2 \cdot \sin 2\alpha + \tau_{xy} \cos 2\alpha \; ; \; \tan 2\alpha_p = 2\tau_{xy}/(\sigma_x-\sigma_y)$$

- Mohr's circle: centre at $(\sigma_x+\sigma_y)/2$, radius = τ_{max}. At the crack line $\theta = 0$, $\sigma_1 = \sigma_2 = K/\sqrt{(2\pi r)}$: the circle degenerates into a **point** (equi-biaxial). In plane strain $\sigma_3 = 2\nu\sigma_1$ so the max shear stress is small ⇒ yielding is suppressed ⇒ brittle behaviour.

B. Elastic constants & energy

- E, ν, $\mu = E/2(1+\nu)$, bulk modulus $K_b = E/3(1-2\nu)$, Lamé constant $\lambda = E\nu/[(1+\nu)(1-2\nu)]$. Only two are independent for an isotropic solid.
- Strain-energy density $W = \tfrac{1}{2}\, \sigma_{ij}\, \varepsilon_{ij}$; plane stress: $W = (1/2E)[\sigma_x^2 + \sigma_y^2 - 2\nu\sigma_x\sigma_y + 2(1+\nu)\tau_{xy}^2]$. Near the tip $W \propto 1/r$.
- **Saint-Venant's principle:** effects of load details are local; far-field stress depends only on resultants. **Superposition:** linear elasticity lets us add solutions ($K_{total} = K_a + K_b$ for the same mode).
- Boundary-value problems: traction BC, displacement BC, mixed. Crack faces: $\sigma_y = \tau_{xy} = 0$ (traction-free) for $|x| < a$.

C. Why the inverse-square-root singularity?

Any solution of $\nabla^4\varphi = 0$ with traction-free flanks ($\theta = \pm\pi$) has eigen-expansion $\varphi = r^{\lambda+1} F(\theta)$. Finite displacement/energy at the tip needs $\mathrm{Re}\,\lambda > 0$, and the **lowest admissible** exponent is $\lambda = 1/2 \Rightarrow \sigma \sim r^{\lambda-1} = r^{-1/2}$ (Williams, 1957). Higher terms (r^0, $r^{1/2}$, …) give the T-stress and multi-parameter fields (Unit IV).

4D. Extra Solved Problems

P1. Plane stress, $K_I = 50$ MPa$\sqrt{m}$, $\sigma_{ys} = 400$ MPa. At $r = 0.5$ mm, $\theta = 0$: $\sigma_y = 50/\sqrt{(2\pi\times0.0005)} = $ **892 MPa** $> \sigma_{ys}$ ⇒ elastic solution invalid there; Irwin plastic radius $r_y = (1/2\pi)(K/\sigma_{ys})^2 = (1/2\pi)(0.125)^2 = $ **2.5 mm**; the elastic K-field is valid only well outside r_y.

P2 (strain-gauge K). Measured $\sigma_y = 250$ MPa at $r = 3$ mm, $\theta = 0$ (K-dominant region): $K_I = \sigma_y\sqrt{(2\pi r)} = 250\times\sqrt{(2\pi\times0.003)} = 250\times0.1373 = $ **34.3 MPa$\sqrt{m}$**.

P3 (G_c). Aluminium sheet $G_c = 24$ kJ/m², $E = 72$ GPa, half-crack $a = 15$ mm. $\sigma_f = \sqrt{(E\, G_c/(\pi\, a))} = \sqrt{(72\times10^9\times24000/(\pi\times0.015))} = $ **191 MPa**.

P4 (periodic cracks). Cracks $2a = 20$ mm, pitch $2W = 100$ mm, $\sigma = 100$ MPa. $Y = \sqrt{[(2W/\pi a)\tan(\pi a/2W)]} = \sqrt{[(100/31.42)\tan(0.3142)]} = \sqrt{(3.183\times0.3249)} = 1.017 \Rightarrow K =$

$100\sqrt{(\pi\times0.01)}\times1.017 = 17.72\times1.017 =$ **18.0 MPa√m**.

P5 (splitting force). $P = 5\times10^6$ N/m (per unit thickness) at the centre, $a = 25$ mm: $K = P/\sqrt{(\pi a)} = 5\times10^6/\sqrt{(\pi\times0.025)} =$ **17.8 MPa√m**.

P6. A $2a = 8$ mm crack, $\sigma = 120$ MPa, $K_{Ic} = 45$ MPa√m. $K = 120\sqrt{(\pi\times0.004)} = 13.5$ MPa√m ⇒ safety factor on $K = 45/13.5 =$ **3.3**; $a_c = (1/\pi)(45/120)^2 =$ **44.8 mm**.

4E. Model Answer – Westergaard Derivation in 8 Steps

1. State assumptions: infinite plate, 2a crack on x-axis, biaxial σ, Mode I, plane stress/strain.
2. Airy function $\varphi = \mathrm{Re}\, Z_2 + y\, \mathrm{Im}\, Z_1$ with $Z_1' = Z$, $Z_2' = Z_1 \Rightarrow \nabla^4\varphi = 0$ automatically.
3. Stresses: $\sigma_x = \mathrm{Re}\, Z - y\, \mathrm{Im}\, Z'$, $\sigma_y = \mathrm{Re}\, Z + y\, \mathrm{Im}\, Z'$, $\tau_{xy} = -y\, \mathrm{Re}\, Z'$.
4. Choose $Z = \sigma z/\sqrt{(z^2-a^2)}$; verify B.C.: (i) $|z| \to \infty$ gives σ; (ii) crack faces traction-free.
5. Shift origin to the tip: $z = a + \zeta$, $Z \approx \sigma\sqrt{(a/2\zeta)} = K/\sqrt{(2\pi\zeta)}$, $K = \sigma\sqrt{(\pi a)}$.
6. Write $\zeta = r\, e^{i\theta}$, separate real & imaginary parts ⇒ σ_x, σ_y, τ_{xy} (boxed results).
7. Integrate Z for displacements u, v (with κ).
8. Conclude: $\sigma_{ij} = K/\sqrt{(2\pi r)}\, f_{ij}(\theta)$; K = single parameter; $G = K^2/E'$.

Quick MCQs

1) Airy stress function satisfies : **$\nabla^4\varphi = 0$**. 2) Westergaard's Z_1 is analytic **except at the crack tips** (branch points $z = \pm a$).

3) For plane strain $E' =$ **$E/(1-\nu^2)$**. 4) Order of singularity of σ at a crack tip : **$r^{-1/2}$**. 5) K–G relation : **$G = K^2/E'$**.

4F. Angular Variation of the K-field

Let $A = K_I/\sqrt{(2\pi r)}$. Values of the angular functions (plane stress/strain, Mode I):

θ (deg)	σ_x / A	σ_y / A	τ_{xy} / A
0	1.000	1.000	0.000
30	0.789	1.143	0.177
60	0.433	1.299	0.000
90	0.354	1.061	-0.354
120	0.500	0.500	-0.433
150	0.436	0.082	-0.177
180	0.000	0.000	0.000

- σ_y is largest on the crack line (θ = 0) and falls to 0 on the faces (θ = 180°) – free

surface. $\sigma_x \to 0$ at 180° too.

- **Max shear stress** occurs at $\theta \approx \pm 70°$ (plane stress); in plane strain it is much smaller because $\sigma_3 \approx 2\nu A$ is close to σ_1 – this is why plane-strain plastic zones are small and toughness is low.
- The **tangential (hoop) stress** for pure Mode I is $\sigma_{\theta\theta} = (K_I/\sqrt{(2\pi r)})\cos^3(\theta/2)$ (used for the mixed-mode kink angle, Unit V); its maximum is at $\theta = 0 \Rightarrow$ the crack grows straight ahead (self-similar).

A. Derivation of displacements (sketch)

1. Strains from Hooke's law (plane stress): $\varepsilon_x = (\sigma_x - \nu\sigma_y)/E$, $\varepsilon_y = (\sigma_y - \nu\sigma_x)/E$.
2. Integrate $\varepsilon_x = \partial u/\partial x$, $\varepsilon_y = \partial v/\partial y$ using the Westergaard stresses.
3. Determine rigid-body terms by requiring symmetry: $v(x, 0) = 0$ ahead of the tip and $u(0, y) = 0$ reference.
4. Result in compact form using Z_1: $2\mu u = ((\kappa-1)/2)\,\mathrm{Re}\,Z_1 - y\,\mathrm{Im}\,Z$; $2\mu v = ((\kappa+1)/2)\,\mathrm{Im}\,Z_1 - y\,\mathrm{Re}\,Z$.
5. Insert $Z_1 = K\sqrt{(2\zeta/\pi)}$, $\zeta = r\,e^{i\theta} \Rightarrow u, v$ given in Section 3.

B. Westergaard vs Williams vs Irwin approach

Approach	Idea	Result / use
Westergaard (1939)	Single analytic function $Z(z)$ for symmetric problems	Closed form K for infinite-plate cases
Irwin (1957)	Used Westergaard near tip, defined K; $G = K^2/E'$	Founded LEFM; K-based design
Williams (1957)	Eigenfunction series $r^{n/2}$ for wedge with free flanks	Gives K, T-stress and higher-order terms
Muskhelishvili	Two complex potentials (general plane problem)	All 3 modes, finite bodies, holes

C. Practice Questions with Answers

1. At $r = 1$ mm directly ahead of a crack with $K_I = 25$ MPa$\sqrt{m}$, find σ_y. Ans: $25/\sqrt{(2\pi\times0.001)} =$ **315 MPa**.
2. Find the critical stress for a $2a = 30$ mm crack in glass ($E = 70$ GPa, $G_c = 8$ J/m²), plane stress. Ans: $\sqrt{(E\,G_c/(\pi a))} = \sqrt{(70\times10^9\times8/(\pi\times0.015))} =$ **3.45 MPa**.
3. Find G and K for $\sigma = 80$ MPa, $2a = 60$ mm, $E = 200$ GPa, $\nu = 0.3$ (plane strain). Ans: $K = 80\sqrt{(\pi\times0.03)} =$ **24.6 MPa$\sqrt{m}$**; $G = (1-\nu^2)K^2/E = 0.91\times(24.6\times10^6)^2/(200\times10^9) =$ **2.74 kJ/m²**.
4. If K_I doubles, what happens to the crack-tip stress at fixed r and to G? Ans: stress

doubles; G quadruples ($\propto K^2$).

5. For $K_I = 30$ MPa$\sqrt{m}$ ($E = 70$ GPa, plane stress) what is the CTOD at $r = 0.5$ mm behind the tip? Use $\delta = 2v = (\kappa+1)K/\mu\ \sqrt{(r/2\pi)}$ with $\kappa+1 = 4/(1+\nu)$. Ans ($\nu = 0.33$): $\mu = 26.3$ GPa, $\kappa+1 = 3.01 \Rightarrow \delta = 3.01\times30\times10^6/(26.3\times10^9) \times \sqrt{(0.0005/2\pi)} = 3.43\times10^{-3} \times 0.00892 = \mathbf{3.1\times10^{-5}\ m \approx 31\ \mu m}$.

5. Summary & Exam Pointers

Quantity	Expression (Mode I)
Westergaard function	$Z_I = \sigma z/\sqrt{(z^2-a^2)}$ (centre crack, biaxial)
Stresses	$\sigma_x = \mathrm{Re}\ Z - y\ \mathrm{Im}\ Z'$; $\sigma_y = \mathrm{Re}\ Z + y\ \mathrm{Im}\ Z'$; $\tau_{xy} = -y\ \mathrm{Re}\ Z'$
Tip field	$\sigma_{ij} = K/\sqrt{(2\pi r)}\ f_{ij}(\theta)$, $K = \sigma\sqrt{(\pi a)}$
Displacements	$\propto K\sqrt{(r/2\pi)}\ g_{ij}(\theta)/(2\mu)$, $\kappa = 3-4\nu$ or $(3-\nu)/(1+\nu)$
Crack opening	$v = (2\sigma/E')\sqrt{(a^2-x^2)}$
K–G	$G = K^2/E'$

1. Derive the stress field near a crack tip for Mode I using the Westergaard stress function.
2. Explain the Airy stress function. Why does $\nabla^4\varphi = 0$?
3. Derive $G = K^2/E'$ (plane stress) using Irwin's crack-closure integral.
4. State the boundary conditions satisfied by Westergaard's function for a centre crack in an infinite plate.
5. Explain plane stress / plane strain and the corresponding values of E', κ.
6. Sketch crack-opening displacement and show it is elliptical.